

Curriculum Vitae

YASSINE YAAKOUBI

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Research Areas: Learning-augmented optimization under uncertainty; machine learning and combinatorial optimization; operations research; decision-focused learning; AI for decision-making and agentic AI; stochastic and distributionally robust optimization; airport and airline operations; sustainable supply/value chains; climate and sustainability.

Selected Current Research Themes

- Learning-augmented optimization under uncertainty for complex operational systems.
- Digital and cyber-physical twins, delay propagation and aircraft-turnaround reliability, airport electrification, and net-zero airport operations.
- Decision-focused learning, robust predict-then-optimize, and ML-augmented solvers.
- Agentic AI, retrieval-augmented reasoning, and evidence-certified decision support.
- Sustainable mining, mineral value chains, and climate-resilient infrastructure.

Education

2017–2020	Ph.D. in Mathematics Polytechnique Montréal, Canada Thesis Title: "Combining Artificial Intelligence and Mathematical Programming for Airline Crew Scheduling" Advisors: François Soumis and Simon Lacoste-Julien Note: Fast-tracked from Master's to Ph.D. program GPA: 4.0/4.0
2017	Maîtrise (M.Sc. equivalent) in Applied Mathematics Polytechnique Montréal, Canada and Institut Polytechnique de Grenoble, France Advisor: François Soumis Note: Fast-tracked to Ph.D. program; dual degree between Institut Polytechnique de Grenoble and Polytechnique Montréal GPA: 4.0/4.0
2014–2017	Engineering Diploma in Information Technology Institut Polytechnique de Grenoble, France Joint degree between Phelma (École Nationale Supérieure de Physique, Électronique et Matériaux) and Ensimag (École Nationale Supérieure d'Informatique et de Mathématiques Appliquées) Triple Major: Discrete Optimization and Operations Research, Networks, IT Security; Double Minor Graduated with highest honors ("mention très bien")

Professional Appointments

2024–present	Tenure-Track Assistant Professor Concordia University, Montréal, Canada Department of Mechanical, Industrial & Aerospace Engineering (MIAE), Gina Cody School of Engineering and Computer Science
2023–2024	Team Lead – Grand Challenge Initiatives in AI for Climate & Nature

- Climate Change AI and Bezos Earth Fund, USA
Led a landscape assessment of the AI-for-Climate Grand Challenge to identify gaps and formulate recommendations
In collaboration with David Rolnick, Priya L. Donti, and Lynn Kaack; informed the \$100M AI for Climate & Nature Grand Challenge
- 2022–2024 **Postdoctoral Fellow**
McGill University, Montréal, Canada
COSMO (Stochastic Mine Planning Laboratory)
Supervisors: Roussos Dimitrakopoulos, Erick Delage, Yossiri Adulyasak, Emma Frejinger
Project: Integrated Machine Learning and Optimization for Decision Making under Uncertainty
- 2022–2023 **Program Manager**
Canadian Institute for Advanced Research (CIFAR)
Led a landscape assessment, symposium, and strategic initiative on AI for Energy and the Environment (AI4E&E)
- 2020–2022 **Postdoctoral Fellow**
McGill University, Montréal, Canada
COSMO (Stochastic Mine Planning Laboratory)
Supervisor: Roussos Dimitrakopoulos
Developed self-learning meta- and hyper-heuristics for stochastic optimization of mining complexes
Spearheaded the development of COSMO Suite software
Assisted in NSERC grant proposals; supervised MSc and PhD students
Served as CIFAR's official reporter for the Pan-Canadian AI Strategy
- 2017–2019 **Research Assistant**
Polytechnique Montréal, Montréal, Canada
GERAD; Mila (Quebec Artificial Intelligence Institute)
Integrated machine learning and combinatorial optimization for airline crew scheduling
Developed ML-augmented warm-starting and adaptive clustering techniques
- 2015–2016 **Research Engineer**
Institut Polytechnique de Grenoble, Grenoble, France
G-SCOP Laboratory
Supervisor: Gautier Stauffer
Applied stochastic traveling salesman problem to golfing strategy optimization
Developed a Q-learning-based reinforcement learning algorithm
- 2015 **Intern**
General Motors, Strasbourg, France
Optimized automatic transmissions production line and managed supply chain operations
Achieved production goal of 3,000 units within the specified timeline
- 2014–2015 **Research Engineer**
Institut Polytechnique de Grenoble, Grenoble, France
GIPSA-lab (Grenoble Images Speech Signal and Control)
Supervisor: Franck Quaine
Engineered a myoelectric interface for real-time EMG signal analysis and classification
Demonstrated control of a physical pilot arm via a 3D virtual arm

Research Funding

Awarded and/or Completed

- [F1] **Digital Technology Programme**
Principal Investigator. 2025/08-2026/08. \$100,000 USD.
Funding Source: Gates Foundation. Funding Competitive?: Yes.
- [F2] **NSERC Discovery Grant**
Principal Investigator. 2025/5–2030/4. \$215,000 CAD plus \$12,500 CAD Discovery Launch Supplement for early-career researchers (\$227,500 CAD total).

- Project: *Machine Learning and Optimization for Sustainable Decision Support in Complex and Uncertain Systems*.
Funding Source: Natural Sciences and Engineering Research Council of Canada (NSERC). Funding Competitive?: Yes.
- [F3] **Google Research / Cloud Compute Credits Programme**
Principal Investigator. 2024/12–2026/1. \$85,000 USD.
Funding Source: Google Research / cloud compute credits. Funding Competitive?: Yes.
Co-investigators: Amal Rannen-Triki
- [F4] **Faculty Research Start-up Funds**
Principal Investigator. 2024/8–2026/8. \$75,000 CAD.
Funding Source: Concordia University.
- [F5] **Deep Learning Indaba Grant**
Principal Applicant. 2024/8–2025/7. \$34,500 USD.
Funding Source: Schmidt Sciences. Funding Competitive?: Yes.
- [F6] **IVADO Scientist in Residence Program - Intelligent CapEx Optimization for Sustainable Mining**
Collaborator. 2024/8–2025/2. Total Funding: \$10,000 CAD (Scale AI: \$5,000 CAD; IVADO: \$5,000 CAD).
Portion of Funding Received: \$0 CAD. Funding Competitive?: Yes.
Co-applicants: Abdallah Jarray, Luiz Silva, Renaud Sénéchal.
Principal Applicant: Matheus Faria.
- [F7] **Strengthening African Machine Learning and Artificial Intelligence through Deep Learning Indaba**
Co-investigator. 2024/7–2028/6. \$400,000 CAD.
Funding Sources: International Development Research Centre (IDRC), AI4D.
Portion of Funding Received: \$100,000 CAD. Funding Competitive?: Yes.
Co-investigators: Emily Muller, Shakir Mohamed.
- [F8] **Deep Learning Indaba Grant**
Principal Investigator. 2024/5–2025/4. \$138,000 USD.
Funding Source: Bill and Melinda Gates Foundation. Funding Competitive?: Yes.
Co-investigator: Shakir Mohamed.
- [F9] **Integrated Machine Learning and Optimization for Decision Making under Uncertainty**
Co-investigator. 2022/9–2024/8. \$70,000 CAD.
Funding Source: IVADO Strategic Research Funding Program. Funding Competitive?: Yes.
Principal Investigator: Roussos Dimitrakopoulos.
- [F10] **Doctoral Scholarship**
Principal Applicant. 2017/9–2020/2. \$60,000 CAD.
Funding Source: École Polytechnique de Montréal. Funding Competitive?: No.

Under Review

- [UR1] **CRIAQ Exploring-Innovation & NSERC Alliance Advantage – ENVOL: Causal AI, Learning-Augmented Optimization, and Cyber-Physical Twinning for Resilient, Net-Zero Airport Operations**
Lead Principal Investigator. 2026/3–2031/3. \$6,000,000 CAD (total project value including partner cost-share).
Funding Source: Natural Sciences and Engineering Research Council of Canada (NSERC) + Consortium for Research and Innovation in Aerospace in Québec (CRIAQ). Funding Competitive?: Yes.
Partner: Aéroports de Montréal (ADM). Duration: 3 years (CRIAQ) + 5 years (NSERC).
Co-PIs: Ursula Eicker, Nizar Bouguila, Masoumeh Kazemi-Zanjani, Ali Akgündüz.
- [UR2] **FRQNT Établissement de la relève professorale** (New University Researchers)
Principal Investigator. 2026/4–2029/3.
Funding Source: Fonds de recherche du Québec – Nature et technologies. Funding Competitive?: Yes.
- [UR3] **Volt-Age Research Fellowship in Sustainable Electrification**
Applicant. Application submitted 2026/02.

Funding Source: Volt-Age, Concordia University / Canada First Research Excellence Fund. Funding Competitive?: Yes.

[UR4] **Mitacs Accelerate**

Principal Investigator. 2026/1–2028/1. \$200,000 CAD.

Funding Source: Mitacs. Funding Competitive?: Yes.

Partner: Aeroport AI.

[UR5] **An End-to-End Pathway to an AI-Driven Climate-Resilient Africa**

Co-investigator. 2025/6–2028/5. \$600,000 CAD.

Funding Source: International Development Research Centre (IDRC), AI4D. Funding Competitive?: Yes.

Portion of Funding Received: \$100,000 CAD.

Co-investigators: Amal Nammouchi, Brian Halubanza, John Bagiliko, Rendani Mbuva, Sabrina Amrouche, Santiago Hincapie Potes.

Selected Awards and Distinctions

[A1] **Umuntu Award for Services to the African Machine Learning Community**

2024/09. Deep Learning Indaba.

Awarded to individuals or teams demonstrating exceptional, sustained service that strengthens and supports African machine-learning communities.

[A2] **INFORMS Annual Meeting Travel Award**

2022/9–2022/11. INFORMS. \$1,500. Prize/Award.

Related to my role as editor of OR/MS Tomorrow.

[A3] **Deep Learning Indaba Travel Award**

2022/9–2024/9. \$4,000. Prize/Award.

[A4] **IVADO Postdoctoral Research Fellowship**

2022/9–2024/8. \$70,000 CAD. Honor.

Under the IVADO Strategic Research Funding Program.

[A5] **International Conference on Machine Learning Student Volunteer Award**

2021/6–2021/8. Honor.

[A6] **COSMO Consortium Fellowship**

2020/9–2024/9. \$200,000 CAD. Prize/Award.

[A7] **École Polytechnique de Montréal Doctoral Fellowship Award**

2018/1–2020/2. \$50,000 CAD. Prize/Award.

[A8] **Institut polytechnique de Grenoble Highest Distinctions**

2018/1. Distinction.

Graduated with highest honors from Grenoble Institute of Technology.

[A9] **Explo'RA Sup Exchange Student Mobility Scholarship**

2018/1–2018/12. \$5,000 CAD. Prize/Award.

Mobility grant for joint dual diploma between Grenoble Institute of Technology and Polytechnique Montréal.

[A10] **Polytechnique Montréal Master's Scholarship Award**

2017. \$10,000 CAD. Prize/Award.

[A11] **Fondation Grenoble INP Excellence Scholarship**

2016. \$2,500 CAD. Prize/Award.

[A12] **Phelma – Grenoble INP Research Award**

2015. \$1,000 CAD. Prize/Award.

Preprints, Reports, and Manuscripts Under Review

Author's name in bold.

- [P1] Hamed, M., & **Yaakoubi, Y.** *STAC-R: Subspace-Trackd Activation Compression with Residuals*. Under review, NeurIPS 2026.
- [P2] Heidari, O. R., & **Yaakoubi, Y.** *NeoDiplomacy: An Intent-Transparent Dataset for Strategic Deception in LLMs*. Under review, EMNLP 2026.
- [P3] Heidari, O. R., Hamed, M., Azzam, D., & **Yaakoubi, Y.** *HippoRAG-Adaptive: Uncertainty-Driven Adaptive Retrieval for Multi-Hop Question Answering*. Under review, NeurIPS 2026.
- [P4] Heidari, O. R., Hamed, M., Azzam, D., & **Yaakoubi, Y.** *ECSSO: Evidence-Certified Safe Overrides for Multi-Hop Question Answering*. Under review, NeurIPS 2026.
- [P5] Azzam, D., Hamed, M., Heidari, O. R., & **Yaakoubi, Y.** *The Compliance Gap in LLM Safety Evaluation*. Under review, NeurIPS 2026 (Evaluations and Datasets Track).
- [P6] Heidari, O. R., Reid, S., Wang, Y., & **Yaakoubi, Y.** *AGENTIQL: Agentic LLMs with Adaptive Routing for Text-to-SQL*. Under review, COLM 2026.
- [P7] **Yaakoubi, Y.**, & Dimitrakopoulos, R. *Distributionally Robust Warm-Starting for Mineral Supply/Value Chains*. Under review, *INFORMS Journal on Computing*.
- [P8] Del Castillo, F., **Yaakoubi, Y.**, & Dimitrakopoulos, R. *Stochastic Optimization of Mining Complexes Integrating Capital Investments and Operational Alternatives*. Under review, *Annals of Operations Research*.
- [P9] Pereira, P., Courtade, E., Aloise, D., Quesnel, F., Soumis, F., & **Yaakoubi, Y.** *Learning to Branch for the Crew Pairing Problem*. Under review, *Transportation Research Part E*.
- [P10] **Yaakoubi, Y.**, Donti, P. L., Kaack, L. H., Rolnick, D., Dunietz, J., Malik, M., Afolabi, T., Kuhne, P., Ndzimande, O., Chinyamakobvu, M., Gombakomba, R., Chibvongodze, V., Leslie, T., Bjärby, E., & Letuka, T. (2024). *Grand Challenge Initiatives in AI for Climate & Nature: Landscape Assessment and Recommendations*. Climate Change AI. Report.
- [P11] Mbuva, R., **Yaakoubi, Y.**, Bagiliko, J., Hincapie Potes, S., Nammouchi, A., & Amrouche, S. (2024). *Leveraging AI for Climate Resilience in Africa: Challenges, Opportunities, and the Need for Collaboration*. arXiv:2407.05210.

Journal Publications

Author's name in bold.

- [J1] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2025). Decision-focused neural adaptive search and diving for optimizing mining complexes. *European Journal of Operational Research*, 320(3), 699–719.
- [J2] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2023). Learning to schedule heuristics for the simultaneous stochastic optimization of mining complexes. *Computers & Operations Research*, 159, 106349.
- [J3] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2022). A data-driven approach for the simultaneous stochastic optimization of mining complexes. *IFAC-PapersOnLine*, 55(21), 67–72.
- [J4] Tahir, A., Quesnel, F., Desaulniers, G., El Hallaoui, I., & **Yaakoubi, Y.** (2021). An improved integral column generation algorithm using machine learning for aircrew pairing. *Transportation Science*, 55(6), 1411–1429.
- [J5] **Yaakoubi, Y.**, Soumis, F., & Lacoste-Julien, S. (2021). Structured convolutional kernel networks for airline crew scheduling. *International Conference on Machine Learning*. PMLR, 139, 11626–11636.
- [J6] **Yaakoubi, Y.**, Soumis, F., & Lacoste-Julien, S. (2020). Machine learning in airline crew pairing to construct initial clusters for dynamic constraint aggregation. *EURO Journal on Transportation and Logistics*, 9(4), 100020.
- [J7] Soumis, F., **Yaakoubi, Y.**, & Lacoste-Julien, S. (2019). Machine learning → mathematical programming for air crew scheduling. *Proceedings of the Triennial Symposium on Transportation Analysis*.

Refereed Conference & Workshop Papers and Presentations

Author's name in bold.

- [C1] Gomez, J., & **Yaakoubi, Y.** (2026). Winter berth allocation under weather uncertainty. *Optimization Days (JOPT)*, HEC Montréal.
- [C2] Mustafa, M., Belcora, M., & **Yaakoubi, Y.** (2026). Risk-aware buffer placement for aircraft turnarounds under partial observability. *Optimization Days (JOPT)*, HEC Montréal.
- [C3] Khaled Abdelkader, M., Moe-Mackosso, C., & **Yaakoubi, Y.** (2026). Distributionally robust predict-then-optimize via Wasserstein ambiguity sets. *Optimization Days (JOPT)*, HEC Montréal.
- [C4] Gomez, J., & **Yaakoubi, Y.** (2026). State-dependent Benders cuts for stochastic winter berth allocation. *CORS 2026* (Maritime Logistics Optimization session).
- [C5] Khaled Abdelkader, M., Moe-Mackosso, C., & **Yaakoubi, Y.** (2026). Distributionally robust SPO+ for contextual optimization under distribution shift. *CORS 2026* (Managing Risks and Robustness with Optimization session).
- [C6] Heidari, O. R., Reid, S., & **Yaakoubi, Y.** (2025). AGENTIQL: An agent-inspired multi-expert framework for text-to-SQL generation. *NeurIPS 2025 Workshop on Efficient Reasoning*. arXiv:2510.10661. An extended version is under review at COLM 2026 (see [P6]).
- [C7] **Yaakoubi, Y.** (2025). Graph-based learning for modeling delay propagation in airline networks. *Proceedings of the Triennial Symposium on Transportation Analysis (TRISTAN)*.
- [C8] **Yaakoubi, Y.** (2025). Machine learning and optimization for decarbonizing high-emitting industries. *Workshop on Combinatorial Optimization and Data Science*, Centre de recherches mathématiques (CRM). Invited talk.
- [C9] **Yaakoubi, Y.** (2025). Machine learning and optimization for decarbonizing high-emitting industries. *AI+OR/MS Bridge*, AAAI Conference on Artificial Intelligence.
- [C10] **Yaakoubi, Y.**, Soubra, R., Atallah, G., Bedrossian, S., Farhat, L., Issa, J., Manchanda, R., & Rahman, N. (2025). Mathematical programming and data analytics toward sustainable mine planning. *CIM Connect 2025*.
- [C11] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2025). Machine learning for robust warm-starting in mining complex optimization under uncertainty. *CIM Connect 2025*.
- [C12] **Yaakoubi, Y.** (2024). Application-driven machine learning and optimization for decarbonizing high-emitting industries. *Mila Sustainability Reading Group*. Invited talk.
- [C13] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2024). Machine learning for distributionally robust warm-starting in mineral supply/value chains. *International Symposium on Mathematical Programming (ISMP)*.
- [C14] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2024). Distributionally robust warm-starting for mineral supply/value chains. *Optimization Days (JOPT)*.
- [C15] Ezzine, L. N., Bengio, Y., Atanane, A., Boukachab, G., Boussif, O., Mahfoud, M., **Yaakoubi, Y.**, Benabbou, L., Boussioux, L., Mitra, P., Jacquillat, A., Den Hertog, D., Bennis, M., El Housni, O., et al. (2023). Leveraging AI for natural disaster management: Takeaways from the Moroccan earthquake. *NeurIPS 2023 Workshop on AI for Humanitarian Assistance and Disaster Response*. arXiv:2311.08999.
- [C16] **Yaakoubi, Y.** (2023). Optimization and learning for mineral value chains. *Imperial Oil (ExxonMobil)*. Invited industrial presentation.
- [C17] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2023). Integrated machine learning and optimization for the simultaneous stochastic optimization of mining complexes. *INFORMS Annual Meeting*.
- [C18] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2023). Diverse candidate generation for a sustainability-aware stochastic optimization of mining complexes. *COSMO Technical Day*.
- [C19] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2023). Context-aware neural branching & diving strategies for optimizing industrial mining complexes. *CORS / Optimization Days*.

- [C20] **Yaakoubi, Y.**, de Carvalho, J. P., & Cutler, J. (2023). Context-aware smart solvers for optimizing supply/value chains. *GERAD-IVADO Contextual Optimization Workshop*.
- [C21] **Yaakoubi, Y.**, Radi, H., & Dimitrakopoulos, R. (2022). Learning on graphs for mineral asset valuation under supply and demand uncertainty. *NeurIPS 2022 Workshop on Graph Learning for Industrial Applications*. arXiv:2212.03865.
- [C22] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2022). Learning to schedule heuristics for the simultaneous stochastic optimization of mining complexes. *AAAI-22 Workshop on Machine Learning for Operations Research (MLAOR)*.
- [C23] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2022). Rethinking optimizers and continual learning: Combining AI and OR for optimizing mining complexes under uncertainty. *COSMO Technical Day*.
- [C24] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2022). Self-learning hyper-heuristics for the optimization of industrial mining complexes. *Optimization Days (JOPT)*.
- [C25] **Yaakoubi, Y.**, Soumis, F., & Lacoste-Julien, S. (2021). Structured convolutional kernel networks for airline crew scheduling. *Proceedings of the 38th International Conference on Machine Learning (ICML)*, PMLR 139, 11626–11636.
- [C26] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2021). A self-learning hyper-heuristic method for strategic mine planning. *INFORMS Annual Meeting*.
- [C27] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2021). A self-learning tree-based approach to the simultaneous stochastic optimization of mining complexes. *COSMO Technical Day*.
- [C28] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2021). Learn to perturb: A deep reinforcement learning approach to adaptive simulated annealing for optimizing industrial mining complexes. *European Conference on Operational Research (EURO)*.
- [C29] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2021). A self-learning hyper-heuristic method for strategic mine planning. *Conference of the International Federation of Operational Research Societies (IFORS)*.
- [C30] **Yaakoubi, Y.**, & Dimitrakopoulos, R. (2020). Learn to perturb: A deep reinforcement learning approach to adaptive simulated annealing for optimizing industrial mining complexes. *COSMO Technical Day*.
- [C31] Soumis, F., **Yaakoubi, Y.**, & Lacoste-Julien, S. (2019). Machine learning to mathematical programming for air crew scheduling. *Proceedings of the Triennial Symposium on Transportation Analysis (TRISTAN)*.
- [C32] **Yaakoubi, Y.**, Soumis, F., & Lacoste-Julien, S. (2019). Machine learning in airline crew pairing to construct initial clusters for dynamic constraint aggregation. *Optimization Days (JOPT)*.
- [C33] **Yaakoubi, Y.**, Lacoste-Julien, S., & Soumis, F. (2019). Structured convolutional kernel networks for airline crew scheduling. *Montreal AI Symposium*.
- [C34] **Yaakoubi, Y.**, Soumis, F., & Lacoste-Julien, S. (2018). Accelerating the optimization of aircrew rotations with machine learning. *Optimization Days (JOPT)*.

Teaching Experience

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| 2024–present | Tenure-Track Assistant Professor , Concordia University, Gina Cody School of Engineering and Computer Science
INDU 6161 – Design & Operations of Supply Chain Networks (graduate; Fall 2024, Fall 2025)
INDU 6611 – Applied Industrial Systems Analytics (graduate; Fall 2025)
INDU 498 – Data Analytics for Industrial Engineers (undergraduate; developed; Winter 2025, Winter 2026)
MIAE 383 – Applied Machine Learning in Mechanical, Industrial and Aerospace Engineering (designed; approved as a required core course for all three MIAE undergraduate programs; launching Fall 2026) |
| 2022–2024 | Instructor , McGill University, Faculty of Engineering
MIME 522 – Discrete Optimization and Mineral Resources. |

MIME 631 – Advanced Stochastic Optimization in Mine Planning.
 MIME 513 – Mine Planning Optimization Under Uncertainty.

Teaching Effectiveness and Curriculum Development

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| 2025–2026 | Official Concordia course evaluations
Five evaluated course sections, 132 enrolled students, 91 respondents, 68.94% response rate; aggregate means roughly 1.83–2.21 where 1 is best. Student comments emphasize clarity, engagement, detailed feedback, and real-world application. |
| 2025–2026 | MIAE 383 / Applied Machine Learning in Engineering
Developed and advanced a new department-level applied machine learning course for MIAE students; approved as a required core course for a planned Fall 2026 launch. |
| 2025 | Faculty AI Training in Engineering
Contributed written material to the Gina Cody School AI-curriculum effort, including prerequisite structure, course sequencing, responsible AI, and department-level AI training. |
| 2025–present | Smart Airport Operations professional certificate
Co-developed professional training materials connected to airport operations, AI, and decision support. |

Research Affiliations

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| 2024–present | Concordia Institute of Aerospace Design and Innovation (CIADI) , Concordia University |
| 2024–present | Next-Generation Cities Institute , Concordia University |
| 2024–present | Applied AI Institute (AI²) , Concordia University |
| 2020–present | GERAD – Group for Research in Decision Analysis |
| 2020–present | CIRRELT – Interuniversity Research Centre on Enterprise Networks, Logistics and Transportation |
| 2017–present | Mila – Quebec Artificial Intelligence Institute |

Academic Service and Leadership

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| 2026 | Sustainable Airport Areas International Conference (SAAIC 2026)
Organizing Committee Member. Conference planned for Montréal, 2026. |
| 2026–present | GenAI in MIAE Committee , MIAE, Concordia University |
| 2026–present | Continuous Improvement Process - Use of Engineering Tools (UET) Committee
Gina Cody School of Engineering and Computer Science, Concordia University; committee co-chair. |
| 2025–present | SDG Innovation Network (Réseau SDG Innovation) – Concordia Co-Lead
Institutional co-lead for Concordia University within the Smart-Digital-Green network; covers Offensive de transformation numérique (OTN) round 2 activities and liaison with partner institutions. |
| 2025–present | AI for Decision-Making Reading Group (AI4DM-RG)
Founder and organizer. Regular seminar/research community on AI for decision-making. Funded by the Applied AI Institute (AI ²)
ai4dm-rg.github.io
Hosted and coordinated research discussions with invited academic and industry speakers, including researchers from Columbia University, UC Berkeley, and DeepMind. |
| 2025–present | GCS Innovation Fund Advisory Committee
Gina Cody School of Engineering and Computer Science, Concordia University. |
| 2025–2026 | Continuous Improvement Process - Use of Engineering Tools (UET) Committee |

- Gina Cody School of Engineering and Computer Science, Concordia University; committee member/participant.
- 2025–present **Departmental Graduate Studies Committee (DGSC)**
Department of Mechanical, Industrial & Aerospace Engineering, Concordia University.
- 2024–present **Implementation of AI Training in Engineering Working Group**
Gina Cody School of Engineering and Computer Science, Concordia University.
- 2024–present **AfriClimate AI**
Co-founder. Bridging African AI talent with climate action and resilience.
- 2024–present **Deep Learning Indaba**
Steering Committee Member; Sponsorship and Fundraising Chair.
The Deep Learning Indaba is the largest charity for AI education in Africa. Annual programs include more than 250 travel grants, more than 650 accommodation scholarships, and more than 20 innovation grants. As Sponsorship and Fundraising Chair, helped raise more than \$3 million CAD cumulatively to strengthen machine learning and AI in Africa.
- 2022–present **EDI Committee of GERAD** – Co-founder. Comprehensive survey, data analysis, and strategic recommendations.
- 2022–2023 **Deep Learning Indaba** – Sponsorship Committee Member
Contributed to raising more than \$600,000 USD annually to strengthen machine learning in Africa; led Indaba initiatives on optimization and mining.
- 2022 **AAAI Conference on Artificial Intelligence** – Session Co-Chair.
- 2021–present **INFORMS OR/MS Today & OR/MS Tomorrow** – Editorial staff writer and board member.
- 2021 **International Conference on Machine Learning (ICML)** – Session Co-Chair.
- 2021 **Summer Undergraduate Research in Engineering Poster Competition**, McGill University – Evaluator and follow-up mentor.
- 2018–2019 **Optimization Days (JOPT)** – Session organizer.
- 2017–2019 **DeepAlpha Startup** (Reinforcement Learning for Portfolio Optimization) – Technical advisor.

Graduate Thesis Committees (not including own graduate students)

- 2026 **Jiaxun Guo** – Ph.D. Examination, Committee Member, Concordia University.
“Structured Probabilistic Latent Modeling for Representation Learning and 3D Perception.”
- 2026 **Sami Ben Brahim**
PhD Comprehensive Exam - Examiner
Institution: Concordia University
Title: “Challenges in Deep-Learning-Based Time Series Forecasting Models: Data Scarcity and Model Explainability”
Supervisors: Nizar Bouguila and Manar Amayri
- 2025 **Shima Ghaneizare**
PhD Thesis Defense — Committee Member
Institution: Concordia University
Thesis Title: “Mathematical Models and Solution Algorithms for Resilient Network Design”
Supervisor: Claudio Contardo
- 2025 **Fatemeh Safaei**
PhD Comprehensive Exam — Examiner
Institution: Concordia University
Title: “Integrated Assortment and Shelf Space Planning with Considering Machine Learning Approaches”
Supervisors: Masoumeh Kazemi Zanjani and Satyaveer Chauhan
- 2025 **Neetusha Kalarikkal**
PhD Thesis Defense — Arms-Length Examiner
Institution: Concordia University

- 2025 Thesis Title: “Emulation of Faults in Permanent Magnet Synchronous Machines”
Supervisor: Pragasen Pillay
Santiago Valencia Ibañez
PhD Comprehensive Exam — Examiner
Institution: Concordia University
Title: “System Architecture Optimization for Reduced-Crew Operation Aircraft”
Supervisor: Susan Liscouët-Hanke
- 2024 **Sara Mohammadi**
PhD Thesis Committee Member
Institution: Concordia University
Title: “Sustainability and Mathematical Programming-based Supply Chain in Wood Industry”
Supervisor: Mingyuan Chen

Research Supervision and Mentoring

- Current group** 10 graduate researchers at Concordia University: 1 postdoctoral fellow, 7 PhD students, 2 MSc students.
- Incoming (Fall 2026)** 2 PhD students and 1 MSc student.
- Summer 2026 interns** 8 undergraduate research interns.

Postdoctoral Fellows

- 2025–present **Xuan Li** (Concordia University; Supervisor)

Doctoral Students

- 2026–present **Fuqiang (Sky) Lu** (Concordia University; Supervisor; incoming, Fall 2026)
- 2026–present **Abdulaziz Aldoseri** (Concordia University; Supervisor; incoming, Fall 2026)
- 2025–present **Omid Reza Heidari** (Concordia University; Supervisor)
- 2025–present **Manar Hamed** (Concordia University; Supervisor)
- 2025–present **Javier Gomez** (Concordia University; Supervisor)
- 2025–present **Chelvy Moe-Mackosso** (Concordia University; Supervisor)
- 2025–present **Mahmoud Khaled Abdelkader** (Concordia University; Supervisor)
- 2025–present **Xiang Chen Zhu** (Concordia University; Supervisor)
- 2025–present **Diaa Ayman Hamdy Azzam** (Concordia University; Supervisor)
- 2019–2020 **Adil Tahir** (Polytechnique Montréal; Co-supervisor). Present position: Assistant Professor, Mohammedia Faculty of Science and Technology.
- 2019–2020 **Fernanda Del Castillo** (McGill University; Co-supervisor). Present position: Director of AI Governance, Mastercard.

Master's Students

- 2026–present **Youssra Ouled Alla** (M.A.Sc., Concordia University; Supervisor; incoming, Fall 2026)
- 2025–present **Ryan Naccache** (M.A.Sc., Concordia University; Supervisor)
- 2025–present **Mohammad Mustafa** (M.A.Sc., Concordia University; Supervisor)
- 2022–2023 **Cristina Tessa Penadillo Palomino** (M.Eng., McGill University; Co-supervisor). Present position: Long-Term Planner, Newmont.
- 2022 **Hager Radi** (M.Sc., University of Alberta; Co-supervisor). Present position: Senior Applied Research Scientist, Mila.
- 2021–2022 **Pierre Pereira** (M.Sc., Polytechnique Montréal; Co-supervisor). Present position: R&D Engineer, JoliBrain.
- 2021–2022 **Emeric Courtade** (M.Sc., Polytechnique Montréal; Co-supervisor). Present position: Data Scientist, CMA CGM.
- 2021 **Philippe Decoste** (M.Eng., McGill University; Co-supervisor).
- 2017–2018 **Philippe Racette** (M.Sc., Polytechnique Montréal; Co-supervisor). Present position: Ph.D. Student, Polytechnique Montréal.

Graduate Capstone and Research Mentees

- 2024–2025 **Graduate capstone mentees** (M.A.Sc./M.Eng., Concordia University): Sadjad Siadat, Muneeb Ahmad, Mohammad Saad, John Sam Daniel, Manan Rajendra Patel, Abhinav Deshwar, Aditya Thakkar, Divyanshu Jaggi, Om Raval, Yash Patel, Param Patel.
- 2024–2025 **Meryem Belcora** (M.Eng., Concordia University; mentor). Mentored through ENGR 6971 and ENGR 6981.

Undergraduate Research Students

- 2025–2026 **Undergraduate capstone team — Container Recycling Network** (B.Eng., Concordia University; mentor): Marcu Badan, Arab Youva Benighil, Hugo Windsore Dabiré, Joseph Helou, Aditya Shah, Marc Zaarour.
- 2024–2025 **Undergraduate capstone team — Strategic Mining Facility Relocation** (B.Eng., Concordia University; mentor): Georges Atallah, Stephanie Bedrossian, Louwy Farhat, Jad Issa, Prithish Manchanda, Nadia Rahman.
- 2024–2025 **Logan Labossiere** (B.Sc., McGill University; advisor).

Research Interns

All beginning in 2026.

- 2026 **Zié Berthé** (École Polytechnique (X), France). Project: *Zoom-then-Optimize: Multi-resolution Large-Scale Graph Optimization*.
- 2026 **Omar Abdelhedi** (École Polytechnique de Tunisie). Project: *Uncertainty-Driven Adaptive Orchestration in Multi-Agent Retrieval-Augmented QA Systems*.
- 2026 **Sahar Chaari** (École Nationale des Sciences de l'Informatique (ENSI), Tunisia). Project: *Robust Multi-Source Retrieval and Verification for Multi-Agent QA Systems*.
- 2026 **Souleymane Diallo** (Université Amadou Mahtar Mbow, Senegal). Project: *Modular Reinforcement Learning Framework for Electric Vehicle Routing Problems*.
- 2026 **Dimeth Nouicer** (Deggendorf Institute of Technology, Germany). Project: *Beyond Least-Cost: Mapping Cost–Carbon Trade-offs for Regional Energy Transitions*.
- 2026 **Al Hakim Taoufik** (École Polytechnique (X), France). Project: *Physics-Informed Sequential Decision-Making for Airport Operations via Reinforcement Learning*.
- 2026 **Aymen Lassoued** (École Polytechnique de Tunisie). Project: *Meta-Reinforcement Learning for Aviation Operations*.
- 2026 **Yicheng Zhang** (Hong Kong Polytechnic University). Project: *AI-Driven Aviation Operations Optimization*.

Reviewing Experience

- **Journals:** INFORMS Journal on Computing, Resources Policy, Transportation science, Computers & Geosciences, Pattern Recognition, International Journal of Mining Science, Technology, and Deep Learning Indaba.
- **Conferences:** ICML (International Conference on Machine Learning), AISTATS (International Conference on Artificial Intelligence and Statistics), International Conference on Computer Science and Application Engineering, IFAC (International Federation of Automatic Control Symposium on Control), Optimization and Automation in Mining, Mineral, Metal Processing, and JOPT (Optimization Days).
- **Workshops:** Montreal AI Symposium (MAIS).

Certifications & Skills

- **Professional:** Ordre des ingénieurs du Québec (OIQ) – member no. 6063606; professional examination completed October 21, 2023.
- **Programming languages:** Python, C/C++, C#, Java, R.
- **Software libraries and tools:** JAX, PyTorch, TensorFlow, Keras, scikit-learn, Gurobi, CPLEX.
- **Languages:** English (C2), French (C2), Arabic (C2), German (B2).

Media Coverage

- [M1] **The path for AI in poor nations does not need to be paved with billions.** *Nature* (Editorial), May 21, 2025.
- [M2] **Africa leading the global effort for AI that works for all**, by Georgina Curto, Frank Dignum, Girmaw Abebe Tadesse, Avishkar Bhoopchand, Sibusisiwe Makhanya, Vukosi Marivate, and Emma Ruttkamp Bloem. *Nature Africa* (Comment), May 21, 2025.
- [M3] **AI models in the Global South** (Collection), edited by Nicola Nosengo, Subhra Priyadarshini, Sahana Ghosh, Akin Jimoh, Lynne Smit, and Saad Lotfey. *Nature* (Participating journals: *Nature*, *Nature Machine Intelligence*, *Nature India*, *Nature Africa*, *Nature Middle East*), May 21, 2025.
- [M4] **Thinking Ahead: As the Use of Artificial Intelligence and Machine Learning in Mining is Becoming More Commonplace, Research is Underway to Automate Every Part of a Mining Operation.** *CIM Magazine*, February 4, 2022.
- [M5] **Africa's AI Researchers are Ready for Takeoff**, by Melissa Heikkilä. *MIT Technology Review*, November 12, 2024.
- [M6] **What Africa Needs to Do to Become a Major AI Player**, by Abdullahi Tsanni. *MIT Technology Review*, November 11, 2024.

Intellectual Property

Intellectual Property Patent & License Inventor

Simultaneous Stochastic Optimization of Mining Complexes for Strategic Planning First stochastic mine-planning software platforms on the market, developed in collaboration with (and commercialized to) a consortium of several of the world's largest mining companies (including Newmont, BHP, Vale, and Anglo American) that collectively represent 75% of the world's mining activity by production value.
